Exno: 2
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Study of different type of Network cables

AIMS:

To study and understood different types of network cable and to make our own ethernet cross over / straight cable

Theory:

1) Unshielded twisted pair [UTP]
2) Shielded twisted pair [STP]
3) coaxial cable
4) Fibre optical cable

Cable Type	category	Max. Data Transmiss-	Advantage/ Disadvantage	Use
UTP	3	10bps	) Cheap in cost) easy to install) More prone to EMI	) 10base-T ethernet) Fastethernet gigabit ethernet
	5	100mbps		
	5e	1gbps		
STP/ FSTP	6, 6a	10gbps	) Shielded) Fasterthat UP Disadvantage) Expensive	) gigabit ethernet 10gbethernet) Widely used datacent-
	7	10gbps		

Straight thru cable
X-over cable
1 RX+
2 RX-
3 TX+
6
7
8
1 TX+
2 TX-
3 RX+
4
5
6 RX-
7
8

A
B

A
B

coaxible cable	RG-6 RG-58 RG-11	10-100 mbps	Advantages •) High bandwidth •) Immune to Interface •) Loss is low Disadvantage •) cost •) Size bulky	space of small 5 500 m, Television network, High speed, internet connection
Fibre optics cables	single mode multi mode	100gbps	Advantages •) High speed •) High bandwidth •) High security Disadvantage •) Expensive •) Requires installation	Maximum distance of fibre optics is around 100 meters

Results

Thus, the different types of network cable were studied successfully